

Zhuoran Yu

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Education

Ph.D. in Computer Science

Sep 2021 – May 2026 (Expected)

University of Wisconsin–Madison, Madison, WI

Advisor: Yong Jae Lee

M.S. in Computer Science

Aug 2018 – Aug 2020

Georgia Institute of Technology, Atlanta, GA

GPA: 4.0/4.0

B.Math in Computer Science and Applied Mathematics

Jan 2014 – Jun 2018

University of Waterloo, Waterloo, Canada

Graduate with Distinction & Dean's Honor List

Arthur Beaumont Memorial Scholarship

Technical Expertise

Multimodal LLM Training

- One of the initiators of IBM's next-generation Granite Vision model.
- Investigated multimodal data mixture strategies for early-fusion pretraining (Chameleon).

Evaluation & Benchmarking

- Currently leading the development of audiovisual and chart/document understanding benchmarks focusing on tasks that require strong multimodal reasoning and integration.

Publications & Preprints

* denotes equal contribution

Under Review

See, Hear, and Understand: Benchmarking Audiovisual Human Speech Understanding in Multimodal Large Language Models

Le Thien Phuc Nguyen*, **Zhuoran Yu***, Samuel Low Yu Hang, Subin An, Jeongik Lee, Yohan Ban, SeungEun Chung, Thanh-Huy Nguyen, JuWan Maeng, Soochahn Lee, Yong Jae Lee
Under review.

Revisiting Active Speaker Detection: An In-the-Wild Benchmark for Generalization and Robustness

Le Thien Phuc Nguyen*, **Zhuoran Yu***, Khoa Quang Nhat Cao, Yuwei Guo, Tu Ho Manh Pham, Tuan Tai Nguyen, Toan Ngo Duc Vo, Lucas Poon, Soochahn Lee, Yong Jae Lee
Under review.

Large Language Model Teaches Visual Students: Cross-Modality Transfer of Conceptual Knowledge

Thomas Liang*, **Zhuoran Yu***, Yong Jae Lee
Under review.

Tracing the Effects of Multimodal Data Mixture

Zhuoran Yu, Bhargavi Paranjape, Yong Jae Lee, Srini Iyer
Under review.

Published

How Multimodal LLMs Solve Image Tasks: A Lens on Visual Grounding, Task Reasoning, and Answer Decoding 

Zhuoran Yu, Yong Jae Lee

Conference on Language Modeling (COLM), Montreal, Canada, 2025.

LASER: Lip Landmark Assisted Speaker Detection for Robustness 

Le Thien Phuc Nguyen*, Zhuoran Yu*, Yong Jae Lee

IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), Tucson, Arizona, 2026.

CuRe: Cultural Gaps in the Long Tail of Text-to-Image Systems 

Aniket Rege, Zinnia Nie, Mahesh Ramesh, Unmesh Raskar, Zhuoran Yu, Aditya Kusupati, Yong Jae Lee, Ramya Korlakai Vinayak

IEEE/CVF International Conference on Computer Vision (ICCV), Honolulu, Hawaii, 2025.

Diversify, Dont Fine-Tune: Scaling Up Visual Recognition Training with Synthetic Images 

Zhuoran Yu, Chenchen Zhu, Sean Culatana, Raghuraman Krishnamoorthi, Fanyi Xiao, Yong Jae Lee

Transactions on Machine Learning Research (TMLR), 2025. **Featured Certificate.**

Denoising and Selecting Pseudo-Heatmaps for Semi-Supervised Human Pose Estimation 

Zhuoran Yu*, Manchen Wang*, Yanbei Chen, Paolo Favaro, Davide Modolo

IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), Waikoloa, Hawaii, 2024.

InPL: Pseudo-labeling the Inliers First for Imbalanced Semi-Supervised Learning 

Zhuoran Yu, Yin Li, Yong Jae Lee

International Conference on Learning Representations (ICLR), Kigali, Rwanda, 2023.

Group R-CNN for Weakly Semi-supervised Object Detection with Points 

Shilong Zhang*, Zhuoran Yu*, Liyang Liu*, Xinjiang Wang, Aojun Zhou, Kai Chen

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), New Orleans, 2022.

Scale-Equalizing Pyramid Convolution for Object Detection 

Xinjiang Wang*, Shilong Zhang*, Zhuoran Yu, Litong Feng, Wei Zhang

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Virtual, 2020.

Experience

Research Scientist Intern

May 2025 – Aug 2025

IBM Watson AI Lab, Cambridge, MA

- Led the initial effort on Granite-4 Vision development.

Research Scientist Intern

May 2024 – Aug 2024

Meta (FAIR), Language and Communication Team, Seattle, WA

- Explored budget-constrained solutions for data mixture practices in early-fusion multimodal LLMs.

Research Scientist Intern

May 2023 – Aug 2023

Meta, CORE AI Team, Menlo Park, CA

- Improved recognition models with synthetic data from pre-trained diffusion models.

Applied Scientist Intern

May 2022 – Aug 2022

AWS AI Rekognition, Seattle, WA

- Developed semi-supervised learning approaches for 2D human pose estimation.

Research Intern

May 2019 – Dec 2019

SenseTime Research, Shenzhen, China

- Conducted research on efficient deep learning to reduce latency of neural networks.
- Worked on object detection for both industrial and academic applications.